

An effective finite-witness theorem for the lower mode threshold of forests

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Abstract

Write $\mu(F)$ for the mean size of a uniformly random independent set of a forest F , and $\alpha(F)$ for its independence number. Let a_* be the infimum of $\mu(T)/\alpha(T)$ over all finite nonempty trees, and let a_K be the same minimum restricted to trees of order at most K . We prove

$$a_K - \frac{20}{\sqrt{K+1}} \leq a_* \leq a_K.$$

Thus every violation with a positive margin has a finite witness of explicitly bounded order, and a_* is a computable real with an explicit approximation algorithm. The principal estimate is degree-free stability under vertex deletion: $|\mu(F) - \mu(F - S)| \leq \pi(6 + 4\sqrt{2})\sqrt{|V(F)||S|}/4$. We also give self-contained proofs that a_* is the common lower asymptotic global-mode threshold for trees and forests, and that their complete limiting mode spectra equal $[a_*, 2/3]$. An exactly specified 5717-vertex tree proves $a_* < 0.473393$, strictly improving the alternating decorated-path bound. The finite approximation also holds uniformly over all activities $0 < \lambda \leq 1$. The exact value of a_* remains open.

1 Finite determination to any prescribed accuracy

The average independent-set size is an established graph invariant; see, for example, Andriantiana et al. [2]. For a finite forest F , let

$$Z_F(x) = \sum_k i_k(F) x^k, \quad \mu(F) = Z'_F(1)/Z_F(1), \quad \alpha(F) = \deg Z_F.$$

The empty forest has $\mu = \alpha = 0$. All ratios below concern nonempty graphs. Define

$$a_* := \inf_{T \text{ finite nonempty tree}} \frac{\mu(T)}{\alpha(T)}, \quad a_K := \min_{\substack{T \text{ tree} \\ 1 \leq |V(T)| \leq K}} \frac{\mu(T)}{\alpha(T)} \quad (K \geq 1).$$

Every a_K is rational and is computable by a finite enumeration.

Theorem 1 (Effective finite-witness bound). *For every integer $K \geq 1$,*

$$a_K - \frac{\pi(6 + 4\sqrt{2})}{2\sqrt{K+1}} - \frac{1}{K+1} \leq a_* \leq a_K. \tag{1}$$

In particular $a_K - 20/\sqrt{K+1} \leq a_ \leq a_K$.*

Corollary 2 (Finite certificates and computability). *For each integer $m \geq 2$, exhaustive exact computation of a_{m^2-1} produces the certified rational enclosure*

$$a_* \in \left[a_{m^2-1} - \frac{20}{m}, a_{m^2-1} \right].$$

Given rational $\varepsilon > 0$, one can compute an interval of width less than ε containing a_* . A direct algorithm has bit complexity $\exp(O(\varepsilon^{-2} \log(1/\varepsilon)))$ as $\varepsilon \downarrow 0$. If some tree satisfies $\mu(T)/\alpha(T) < c - \varepsilon$, then some tree H with

$$|V(H)| \leq \lceil (20/\varepsilon)^2 \rceil \quad \text{satisfies} \quad \mu(H)/\alpha(H) < c.$$

The finite-witness statement permits loss of the margin ε . It does not assert bounded-size witnesses for an exact inequality with zero margin. The enumeration algorithm is a computability result, not an efficient practical method for high precision.

2 A covariance bound with arbitrary activities at most one

For vertex activities $0 < \lambda_v \leq 1$, assign an independent set I weight $\prod_{v \in I} \lambda_v$. Set $S_v = \mathbf{1}_{\{v \in I\}}$, $p_v = \mathbb{E}S_v$, $s_v^2 = p_v(1 - p_v)$, and $Y_v = (S_v - p_v)/s_v$. All variances s_v^2 are positive.

Lemma 3. *For every forest and every such activity vector,*

$$\|\text{Cov}(Y)\|_2 \leq C_0 := \left(1 - \frac{1}{\sqrt{2}}\right)^{-2} = 6 + 4\sqrt{2}.$$

Consequently for real vectors a, b ,

$$\left| \text{Cov} \left(\sum_v a_v S_v, \sum_v b_v S_v \right) \right| \leq C_0 \left(\sum_v a_v^2 s_v^2 \right)^{1/2} \left(\sum_v b_v^2 s_v^2 \right)^{1/2}. \quad (2)$$

Proof. For each oriented edge $u \rightarrow v$, the cavity occupation odds satisfy

$$R_{u \rightarrow v} = \lambda_u \prod_{z \sim u, z \neq v} (1 + R_{z \rightarrow u})^{-1} \leq 1.$$

The marginal odds are $w_v = p_v/(1 - p_v) = \lambda_v \prod_{u \sim v} (1 + R_{u \rightarrow v})^{-1}$. Adjacent occupations are disjoint, hence their correlation is $c_{uv} = -\sqrt{w_u w_v}$. Fix v and put $x_u = R_{u \rightarrow v}$. Since $w_u = x_u/(1 + R_{v \rightarrow u}) \leq x_u$,

$$\sum_{u \sim v} c_{uv}^2 \leq \lambda_v \frac{\sum_u x_u}{\prod_u (1 + x_u)} \leq \frac{1}{2}. \quad (3)$$

Here $\prod(1 + x_i) \geq 2 \sum x_i$ on $[0, 1]^d$: the difference is multiaffine, and at a cube vertex with j ones it is $2^j - 2j \geq 0$.

Root each component and order parents before children. Let $B_{v,u} = c_{uv}$ when u is the parent of v , and let all other entries be zero. The columns of B have disjoint supports, so $\|B\|_2 \leq 1/\sqrt{2}$ by (3). Define $\epsilon = (I - B)Y$. For a nonroot v with parent u , conditional independence across u and the binary nature of S_u give

$$\mathbb{E}(Y_v \mid \text{previous states}) = c_{uv} Y_u.$$

For roots the conditional mean is zero. Therefore the innovations ϵ_v are pairwise orthogonal. Their covariance matrix D is diagonal, with entries 1 at roots and $1 - c_{uv}^2$ elsewhere; in particular $0 \preceq D \preceq I$. Since B is nilpotent,

$$\text{Cov}(Y) = (I - B)^{-1} D (I - B)^{-\top}, \quad \|(I - B)^{-1}\|_2 \leq (1 - 1/\sqrt{2})^{-1}.$$

This proves the norm bound. Applying it to the vectors $(a_v s_v)_v$ and $(b_v s_v)_v$ proves (2). $\square$

3 Deleting vertices costs at most a square root

Theorem 4 (Deletion stability). *If F is a forest of order n and $S \subseteq V(F)$, then*

$$|\mu(F) - \mu(F - S)| \leq C_D \sqrt{n|S|}, \quad C_D = \frac{\pi C_0}{4} < \frac{66}{7}. \quad (4)$$

Proof. The case $S = \emptyset$ is immediate. Set $s = |S| > 0$. At time $t \geq 0$, give each vertex in S activity e^{-t} and every other vertex activity 1. Let $X = \sum_v S_v$, $X_S = \sum_{v \in S} S_v$, and $M(t) = \mathbb{E}_t X$. Differentiation of the finite partition function gives the exact identity

$$M'(t) = -\text{Cov}_t(X, X_S).$$

As $t \rightarrow \infty$, the law converges to the uniform independent-set law on $F - S$, so $M(0) = \mu(F)$ and $M(\infty) = \mu(F - S)$.

For all vertices, $s_v^2 \leq 1/4$. For $v \in S$, the marginal odds are at most e^{-t} , so $p_v \leq e^{-t}/(1 + e^{-t}) \leq 1/2$, and hence

$$s_v^2 \leq \frac{e^{-t}}{(1 + e^{-t})^2}.$$

Applying Lemma 3 throughout the interpolation yields

$$|M'(t)| \leq \frac{C_0}{2} \sqrt{ns} \frac{e^{-t/2}}{1 + e^{-t}}.$$

This bound is integrable and

$$\int_0^\infty \frac{e^{-t/2}}{1 + e^{-t}} dt = 2 \int_0^1 \frac{du}{1 + u^2} = \frac{\pi}{2}.$$

Integration proves (4). Finally $C_0 < 12$ and $\pi < 22/7$ give $C_D < 66/7$. $\square$

Lemma 5 (Small components after few vertex deletions). *For every forest F of order n and integer $K \geq 1$, there is $S \subseteq V(F)$ such that $|S| \leq n/(K + 1)$ and every component of $F - S$ has order at most K .*

Proof. Root each component. In a current rooted component of order greater than K , choose a deepest vertex whose descendant subtree has order greater than K . Every child subtree of this vertex has order at most K . Put the vertex in S , retain its child subtrees as final components, and continue in the part toward the root. Charge this deletion to the vertex and all its child subtrees. Every charge contains at least $K + 1$ vertices, and distinct charges are disjoint. This proves the bound on $|S|$; when the process ends, the remaining rooted components also have order at most K . $\square$

Proof of Theorem 1. Take any nonempty tree T of order n and a separator S from Lemma 5; write $s = |S|$. Additivity over the components of $T - S$ gives

$$\mu(T - S) \geq a_K \alpha(T - S) \geq a_K (\alpha(T) - s).$$

Theorem 4 therefore implies

$$\frac{\mu(T)}{\alpha(T)} \geq a_K - \frac{a_K s}{\alpha(T)} - \frac{C_D \sqrt{ns}}{\alpha(T)}.$$

Since $\alpha(T) \geq n/2$ and $a_K \leq 1/2$ (the singleton tree is allowed),

$$\frac{\mu(T)}{\alpha(T)} \geq a_K - \frac{s}{n} - 2C_D \sqrt{\frac{s}{n}} \geq a_K - \frac{1}{K + 1} - \frac{2C_D}{\sqrt{K + 1}}.$$

Take the infimum over T . The opposite bound $a_* \leq a_K$ is immediate. Finally, $1/(K + 1) \leq 1/\sqrt{K + 1}$ and $2C_D + 1 < 139/7 < 20$ prove the simplified bound. $\square$

Proof of Corollary 2. Substitute $K = m^2 - 1$. Choose an integer $m > 20/\varepsilon$ for the algorithm. To enumerate all tree shapes up to order K , it suffices to enumerate parent arrays $p_v \in \{0, \dots, v-1\}$, $1 \leq v < n$, for each $n \leq K$. Every unlabelled tree admits such a labelling, by rooting it and labelling parents before children. Duplicates cause no difficulty. There are $(n-1)!$ arrays of order n . Integer tree recursions compute $Z_T(1)$, $Z'_T(1)$ and $\alpha(T)$ in polynomial bit complexity: all counts have $O(n + \log n)$ bits. Exact rational comparison then yields a_K . The resulting total cost is $\exp(O(K \log K))$.

For the witness assertion, choose $K = \lceil (20/\varepsilon)^2 \rceil$. The error $20/\sqrt{K+1}$ is strictly less than ε . If $a_* < c - \varepsilon$, then $a_K < c$. A tree attaining a_K is the required witness. $\square$

Corollary 6 (Uniform approximation of the entire activity profile). *For $0 < \lambda \leq 1$, put*

$$\mu_\lambda(T) = \lambda Z'_T(\lambda)/Z_T(\lambda), \quad a_*(\lambda) = \inf_T \frac{\mu_\lambda(T)}{\alpha(T)}, \quad a_K(\lambda) = \min_{1 \leq |V(T)| \leq K} \frac{\mu_\lambda(T)}{\alpha(T)}.$$

Then the same finite family of trees satisfies

$$0 \leq a_K(\lambda) - a_*(\lambda) \leq \frac{20}{\sqrt{K+1}} \quad \text{simultaneously for every } 0 < \lambda \leq 1.$$

With both functions set to zero at $\lambda = 0$, this is a uniform approximation on $[0, 1]$ by finite lower envelopes of explicitly computable rational functions.

Proof. Start the deletion interpolation with all activities equal to λ , and suppress those in S to λe^{-t} . All activities remain at most one; the bound for suppressed vertices remains $s_v^2 \leq e^{-t}/(1 + e^{-t})^2$. Thus the same deletion estimate applies. Component additivity is unchanged, and the singleton gives $a_K(\lambda) \leq \lambda/(1 + \lambda) \leq 1/2$. The proof of Theorem 1 now applies verbatim and uniformly. At zero, $\mu_\lambda(T)/\alpha(T) \leq 2\lambda$, so the stated extension is continuous. $\square$

4 Connection with global modes and the complete limit spectrum

Lemma 7. *Every global mode m of an n -vertex forest satisfies*

$$|m - \mu(F)| < 6n^{3/4}, \quad \left| \frac{m}{\alpha(F)} - \frac{\mu(F)}{\alpha(F)} \right| < 12n^{-1/4}.$$

Proof. Lemma 3 gives $\text{Var } |I| \leq C_0 n/4 < 3n$. Chebyshev's inequality puts at least $3/4$ of the mass in an interval containing fewer than $8\sqrt{n}$ integers. Every global mode thus has probability at least $3/(32\sqrt{n})$. Its contribution to the variance gives $|m - \mu|^2 < 32n^{3/2}$. Divide by $\alpha(F) \geq n/2$ for the second bound. $\square$

Theorem 8 (Common threshold and no gaps). *Let $A_{\mathcal{C}}(n)$ be the minimum of $m/\alpha(G)$ over graphs G of order exactly n in class $\mathcal{C}$ and all their global modes. Then*

$$\lim_{n \rightarrow \infty} A_{\text{tree}}(n) = \lim_{n \rightarrow \infty} A_{\text{forest}}(n) = a_*.$$

The sets of all subsequential limits of normalized global modes, as graph orders tend to infinity, are identical for trees and forests and equal $[a_, 2/3]$.*

Proof. For components H_i of F , the ratio $\mu(F)/\alpha(F)$ is the $\alpha(H_i)$ -weighted average of their ratios; it is therefore at least a_* . Lemma 7 proves the lower limit bound.

For a fixed rooted tree (H, v) of order h , write $n - 1 = rh + s$, $0 \leq s < h$. Join a new hub to the roots of r disjoint copies of H and attach s new leaves to it. The resulting tree T_n has polynomial

$$Z_{T_n}(x) = (1 + x)^s Z_H(x)^r + x Z_{H-v}(x)^r.$$

With $\theta = Z_{H-v}(1)/Z_H(1) < 1$, hub occupation has probability at most θ^r . Consequently

$$|\mu(T_n) - r\mu(H) - s/2| \leq n\theta^r, \quad r\alpha(H) + s \leq \alpha(T_n) \leq r\alpha(H) + s + 1.$$

Thus every mode of T_n , normalized by $\alpha(T_n)$, converges to $\mu(H)/\alpha(H)$. Infimizing over H proves the two limits over all integer orders.

For the upper endpoint, let $e(I)$ count vertices that can be added to an independent set I in a forest. The remaining induced forest is bipartite, so $e(I) \leq 2(\alpha(F) - |I|)$. Extension double counting gives $\mathbb{E}e(I) = \mu(F)$, whence $\mu(F) \leq 2\alpha(F)/3$. The edge K_2 has ratio exactly $2/3$.

Fix $r \in (a_*, 2/3)$ and a tree H with $u = \mu(H)/\alpha(H) < r$. Unions of p copies of H and q copies of K_2 have ratios

$$\frac{p\alpha(H)u + (2/3)q}{p\alpha(H) + q}.$$

Choose integer multiplicities tending to infinity so that these ratios tend to r .

To turn any such forest F of order n into a tree, choose one vertex in each component, join them to a new hub, and give the hub $L = \lceil 3 \log_2(n + 1) \rceil$ new leaves. If the chosen vertices form S , the exact polynomial is

$$Z_T(x) = (1 + x)^L Z_F(x) + x Z_{F-S}(x).$$

Hub occupation has probability at most 2^{-L} , and conditional on its absence the law on F is unchanged. Hence

$$|\mu(T) - \mu(F) - L/2| \leq (n + L + 1)2^{-L}, \quad \alpha(F) + L \leq \alpha(T) \leq \alpha(F) + L + 1.$$

The normalized means, and therefore normalized modes, have the same limit. The endpoint $2/3$ follows by using edges only; a_* follows from the already established minima over every order. The universal bounds exclude all other limits. $\square$

The upper endpoint $2/3$ is known, and the equality question for lower tree and forest peak thresholds is stated in Section 9 of Fang et al. [1]. Our statement uses global modes. Their strict central log-concavity together with strict versions of their initial and terminal extension inequalities identifies their weak local peaks with global modes for sufficiently large forests. This identification is not needed for any of the preceding proofs.

5 An exact witness below the decorated-path bound

Define rooted trees $H_0, \dots, H_7$ as follows: H_0 is one vertex, and H_j is a new root attached to the roots of b_j copies of H_{j-1} , where

$$(b_1, \dots, b_7) = (2, 2, 3, 4, 4, 4, 4).$$

Then $|V(H_7)| = 5717$ and $\alpha(H_7) = 3908$. The following entirely integral recursions specify its ratio exactly. At each stage let A, B count independent sets with root absent and present, respectively; let U, V be the corresponding sums of their cardinalities; and let a_0, a_1 be the largest cardinalities in the two states. Initialize

$$(A, B, U, V, a_0, a_1, n) = (1, 1, 0, 1, 0, 1, 1).$$

For a branching factor b , replace these simultaneously by

$$\begin{aligned} A' &= (A + B)^b, & B' &= A^b, \\ U' &= b(U + V)(A + B)^{b-1}, & V' &= A^b + bUA^{b-1}, \\ a'_0 &= b \max(a_0, a_1), & a'_1 &= 1 + ba_0, \\ n' &= 1 + bn. \end{aligned}$$

Writing $P = U + V$ and $Q = 3908(A + B)$ after the last step gives

$$\frac{\mu(H_7)}{\alpha(H_7)} = \frac{P}{Q} = 0.4733925104219515\dots < \frac{473393}{10^6}.$$

The strict rational upper bound is certified by the positive integer $473393Q - 10^6P$. The strict improvement over $c = \frac{5}{8}(1 - 1/\sqrt{17})$ is certified without floating point by

$$5Q - 8P > 0, \quad 17(5Q - 8P)^2 > 25Q^2.$$

The accompanying script computes these integers from the seven stated updates and checks their signs. It also constructs all 5717 vertices and independently evaluates the ordinary tree recursion to check the compressed construction. Thus

$$a_* < 0.473393 < \frac{5}{8}(1 - 1/\sqrt{17}).$$

The alternating decorated path remains a valid earlier upper bound but is not extremal. This finite witness is not claimed to be optimal either.

Status and limitations

The central additional result here is the explicit modulus in Theorem 1, obtained by combining inhomogeneous covariance control, continuous activity suppression, and a vertex separator. Finite computations alone do not prove that theorem; the universal argument is given in Sections 2–4. The included small-instance audits supplement the written proof and are not a proof-assistant certificate. Exhaustive enumeration at a size sufficient for a useful narrow enclosure has not been performed. The exact value of a_* , an extremal classification, and the extremal density of log-concavity failures are not determined. This manuscript is unreviewed, and publication priority for its statements has not been established.

References

- [1] E. X. Fang, J. Lu, E. Nevo, Y. Yao, and H. Zheng, *Unimodality of Independence Polynomials for Sufficiently Large Forests*, arXiv:2609.20961v1 (2026). <https://arxiv.org/abs/2609.20961>.
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